2.4 模型泛化 Generalization

是骡子是马，拉出来遛遛

随着互联网的普及、人工智能的应用，越来越多的问题能很快得到答案。

——节选自2024高考作文题目

感知世界

回归

分类

聚类

创造（虚拟）世界

文本生成

图像生成

视频生成

语音生成

改变（物理）世界

具身智能

人形机器人

无人驾驶

$f(x)$能解决哪些问题?

回归

Regression

$$f(x)$$

交易历史

拟合发展规律
预测数值

量化交易

找到房屋信息与房价间的联系

f(

学区房	湖景房	商业中心	医疗条件	经济条件	交通情况	房型结构	房屋面积	……

) = 房屋价格

这就是**回归问题**

特点1：任务是**预测**或者**推断**

特点2：目标是找出输入和输出之间的函数关系

特点3：**输出的是连续数值**

前情提要——构建预测模型的三个步骤

步骤1

一大堆数据

步骤3

选择最优模型

步骤2

一堆模型

$f_1, f_2, \dots, f_n$

《钟点工》2000年春节联欢晚会

蛮力法找出最优解

- **给出w和b“所有”的值**

  $w \in [w_{min}, w_{max}]$

  $b \in [b_{min}, b_{max}]$

- **尝试所有的情况**

```
for w in w_list:
    for b in b_list:
        loss(x, y, w, b)
```

- **选择最优的结果**

$$\boxed{w^*, b^* = arg\min_{w,b} L(w, b)}$$

步骤1

步骤2

步骤3

- 查阅大量中医古籍找出2000+中草药制剂
- 筛选出640可能的治疟药方
- 从200种草药种得到380种提取物
- 经历190失败
- 对虐原虫的抑制率达到100%

目标函数（Objective Function)

指导算法选出最优的预测模型。

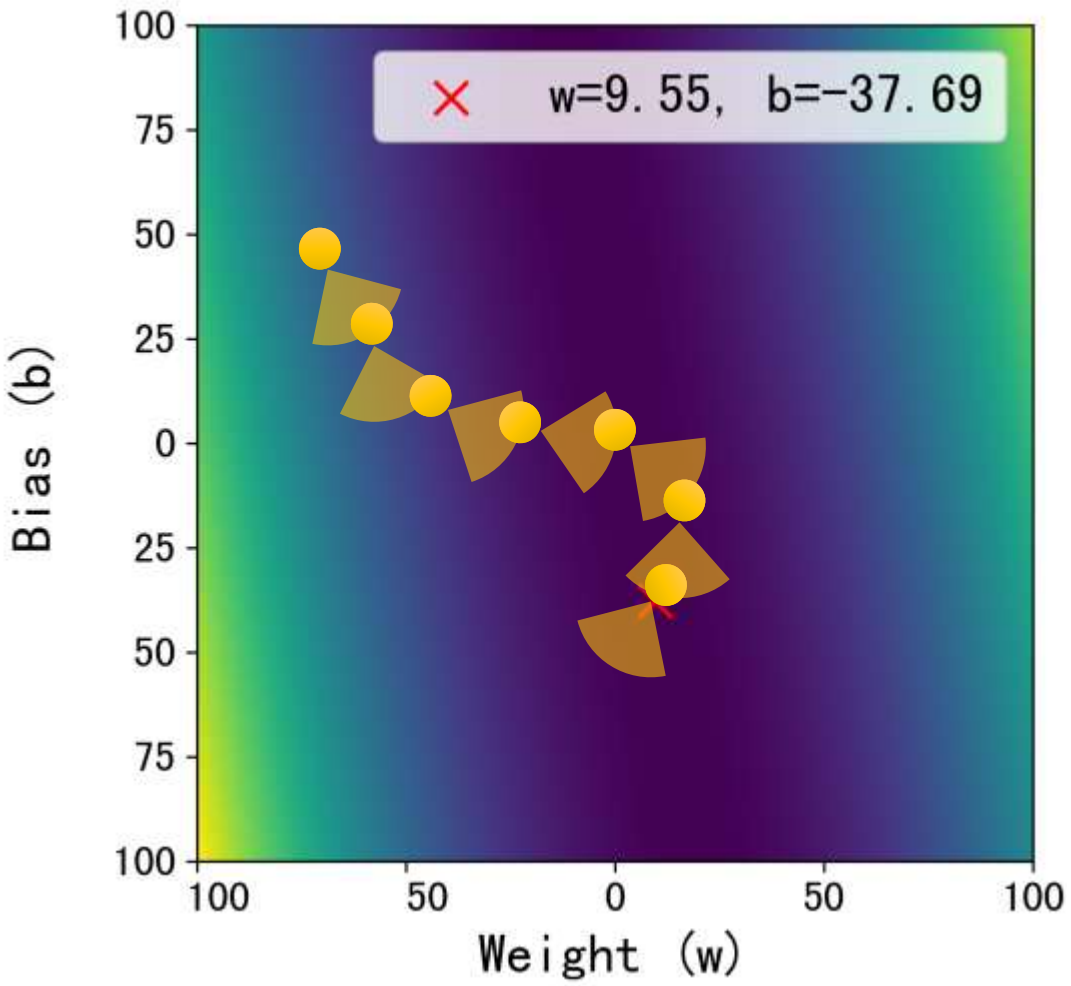
前情提要——梯度下降的基本思想

设想一下：

不论从何处开始，都能判断最优点的大致方向，并快速向其靠近

$$w^*, b^* = arg\min_{w,b} L(w,b)$$

向最小的 $L(w,b)$ 前进



前情提要——使用Gradient Descent方法找到最优的预测模型

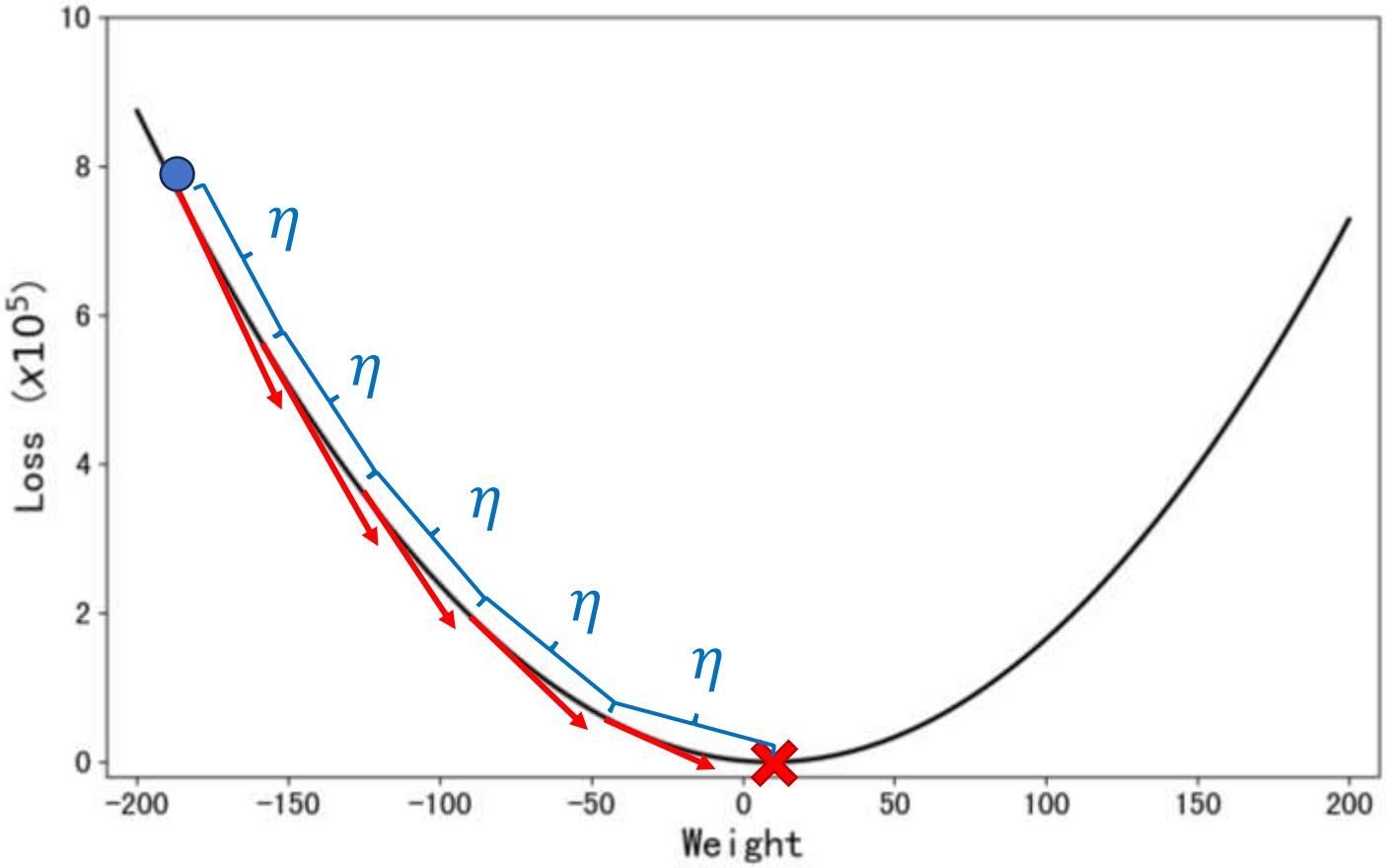
Step 1: 计算各参数的偏导数

$$\frac{\partial L}{\partial b}, \quad \frac{\partial L}{\partial w_i}$$

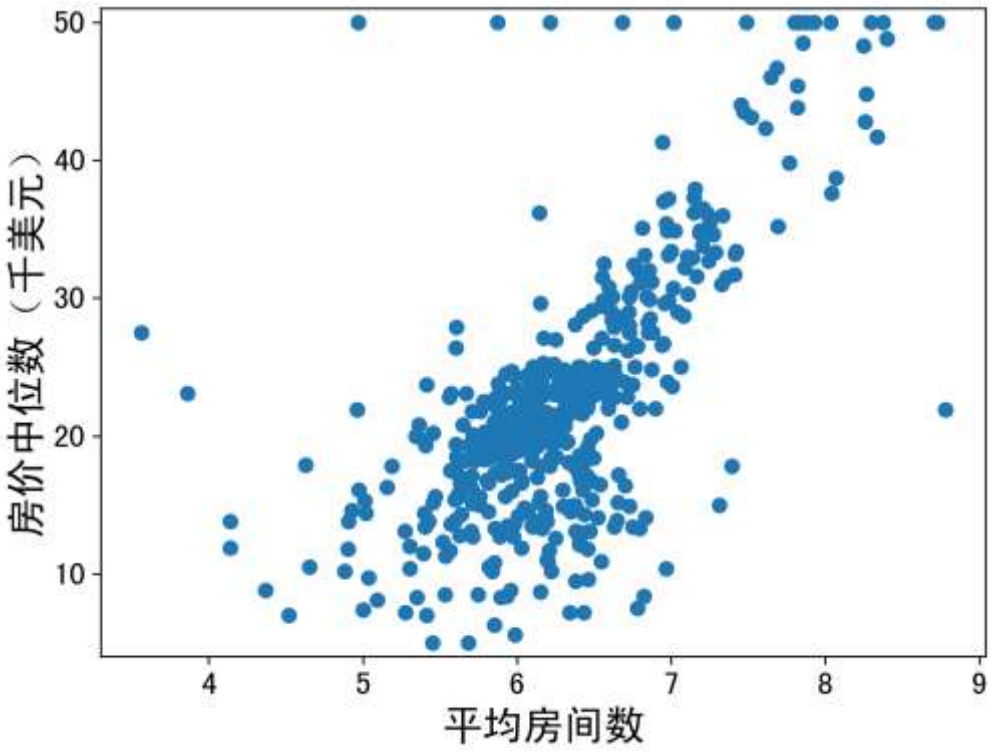
Step 2: 分别更新各参数

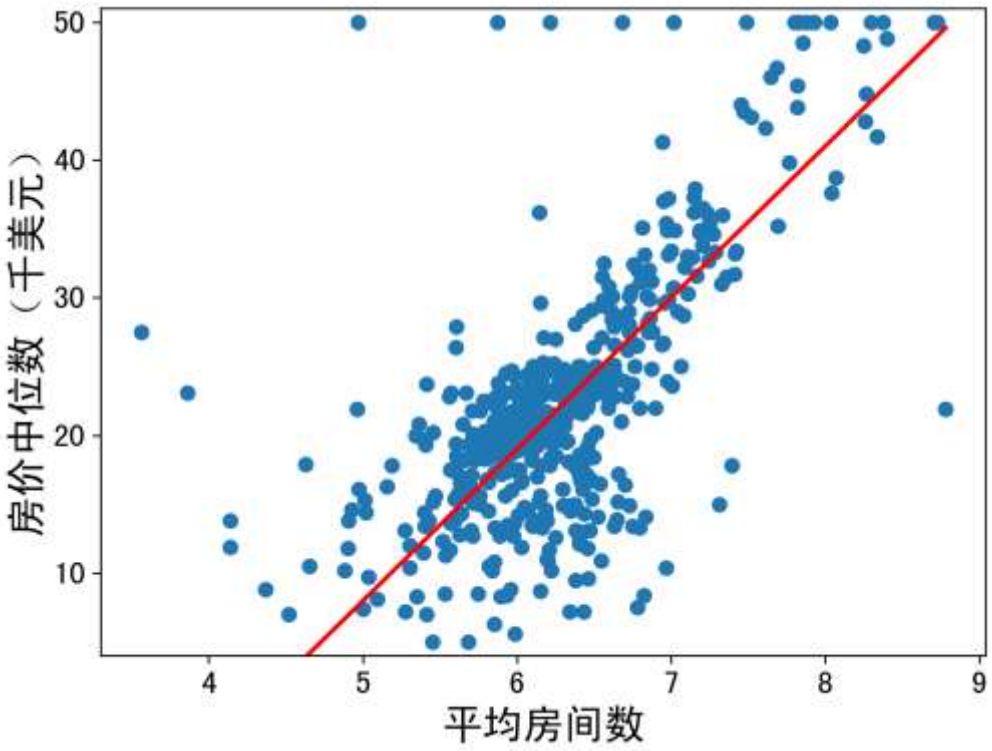
$$b = b - \eta \frac{\partial L}{\partial b}, \quad w_i = w_i - \eta \frac{\partial L}{\partial w_i}$$

Step 3: 重复 Step 1 和 Step 2



思考问题





这样训练完预测模型就能直接用了吗？

泛 化
Generalization

预测模型在实际应用中的关键能力

学习器在**新样本**上的误差称为“泛化误差”。显然，我们希望得到泛化误差小的学习器。（机器学习P23）

机器学习中的**泛化能力**（generalization ability）指的是模型在训练数据之外、在未见过的测试数据或真实世界数据上表现的能力。换句话说，泛化能力衡量模型从有限的训练数据中学到的知识是否能够有效地应用到新数据上。

在机器学习中，泛化能力（Generalization）指的是模型对未见过的数据进行预测的能力。一个具有良好泛化能力的模型能够从训练数据中学习到通用的规律，并且能够将这些规律应用到新的、未见过的数据上，从而做出准确的预测。

泛化能力是机器学习模型性能的关键指标之一，它与模型的过拟合（Overfitting）和欠拟合（Underfitting）密切相关：

如何评估预测模型的准确度?

不好描述 √ $$Loss = \frac{1}{2N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$$

容易理解 √ $$prediction\ error = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|$$

泛化能力评估——训练与测试

训练阶段（Training）

步骤1

一大堆数据

步骤3

选择最优模型

步骤2

一堆模型

$f_1, f_2, \dots, f_n$

- 输入：房价特征
- 输出：房价（标签）

- $y = \Sigma\alpha x + \beta$，$x$是特征值
- f由参数α和β定义

平均房间数
6.998
7.147

步骤4

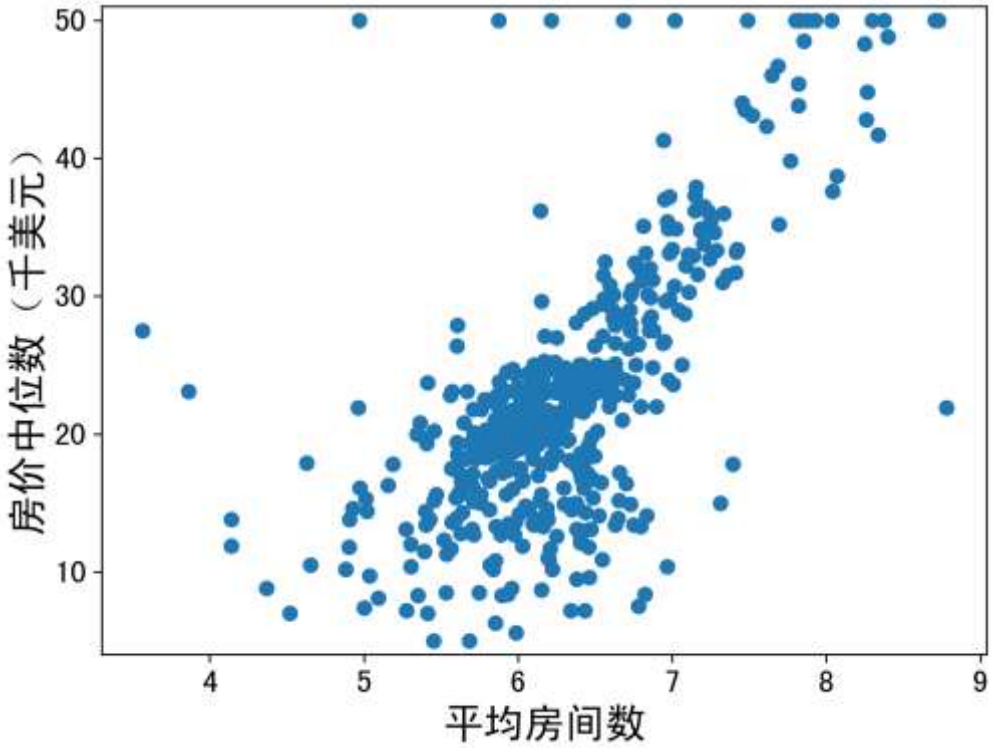
最优模型f_{best}

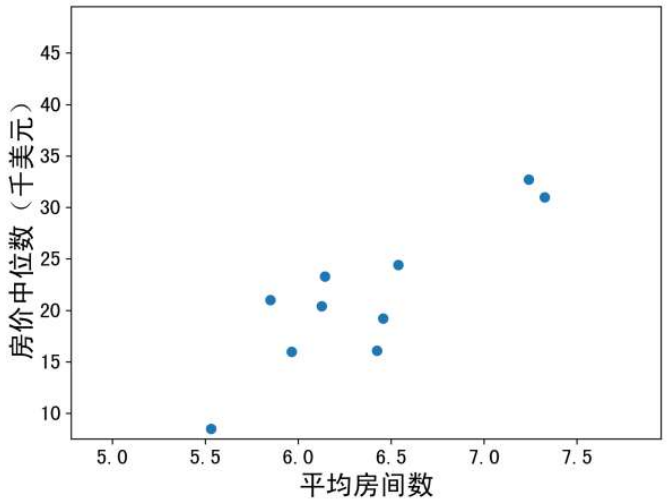
房价（千美元）
33.4
36.2

测试阶段（Testing）

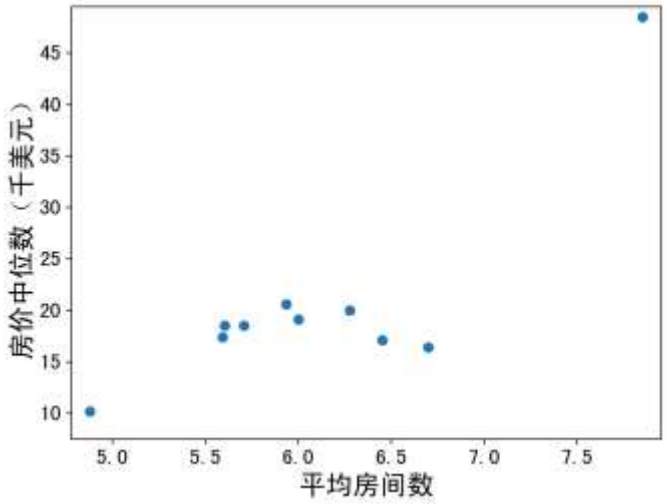
泛化能力评估——训练与测试

Step 1: 分离数据集





训练集



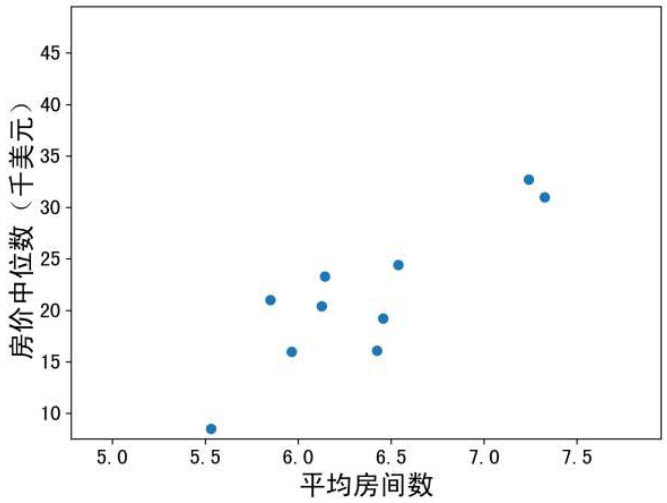
测试集

泛化能力评估——训练与测试

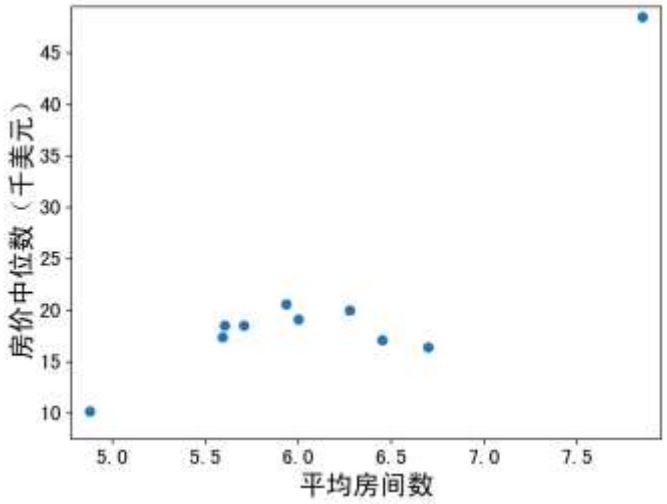
Step 2: 在训练集上训练模型

预测模型

训练



训练集



测试集

泛化能力评估——训练与测试

Step 3: 在测试集上测试模型

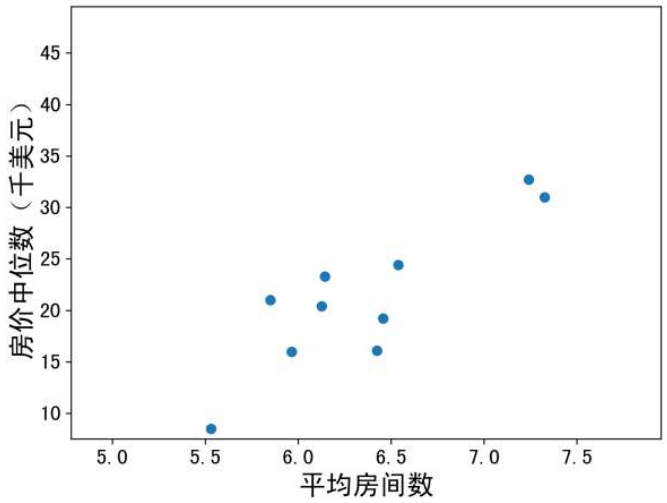
观察训练和测试

结果评估泛化能力

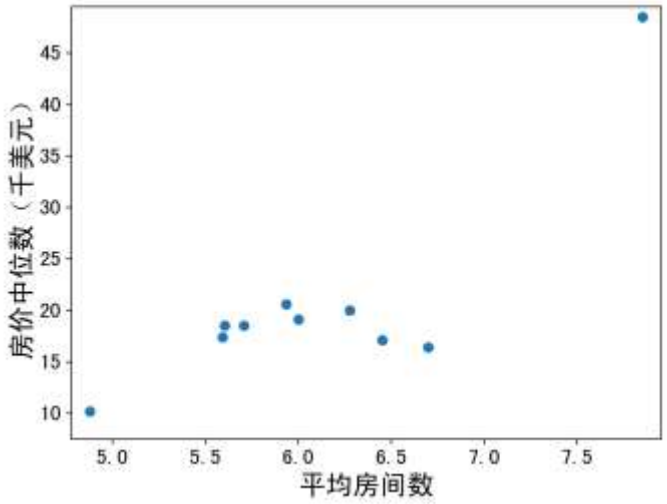
预测模型

训练

测试



训练集



测试集

泛化能力评估——实际案例

Step 1: 分离数据集

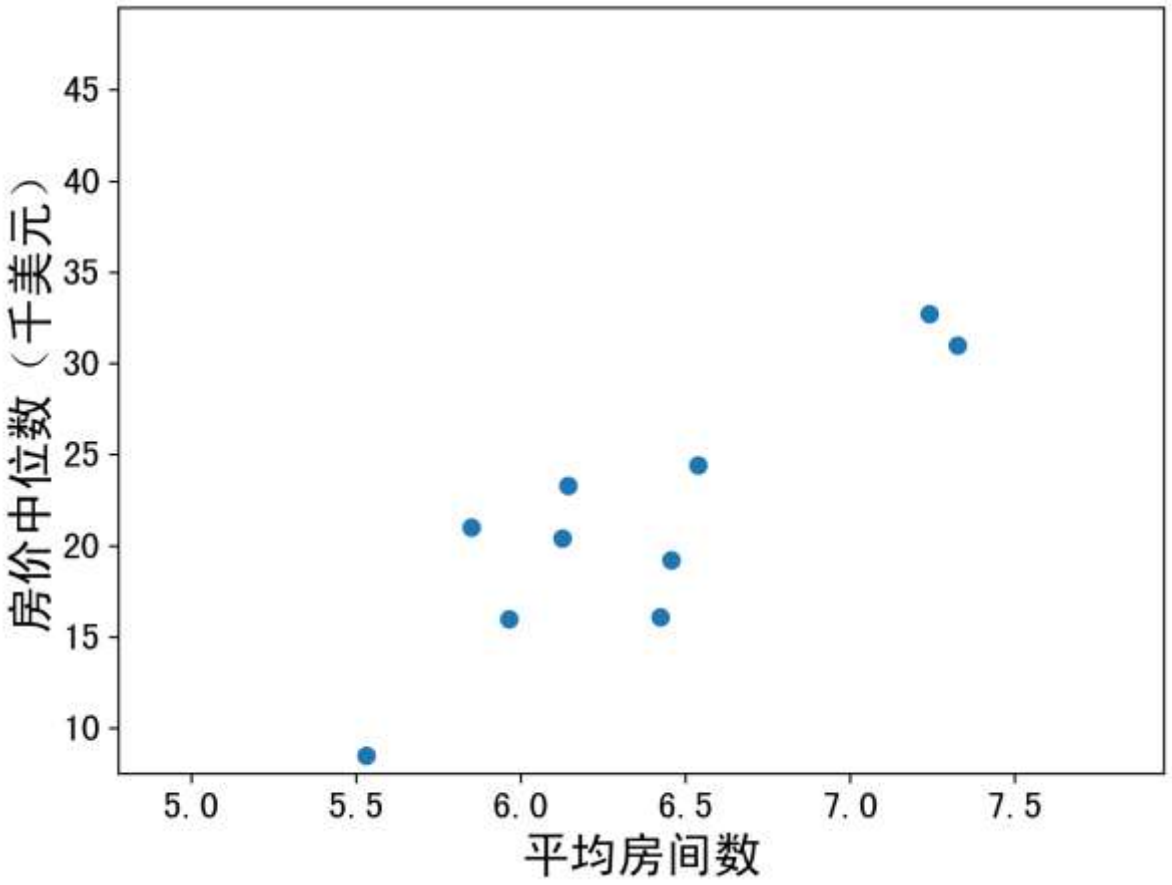
Step 2: 训练模型

- 定义预测函数

$$y = wx + b$$

- 梯度下降拟合模型

Step 3: 测试训练结果



泛化能力评估——实际案例

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Step 2: 训练模型

- 定义预测函数

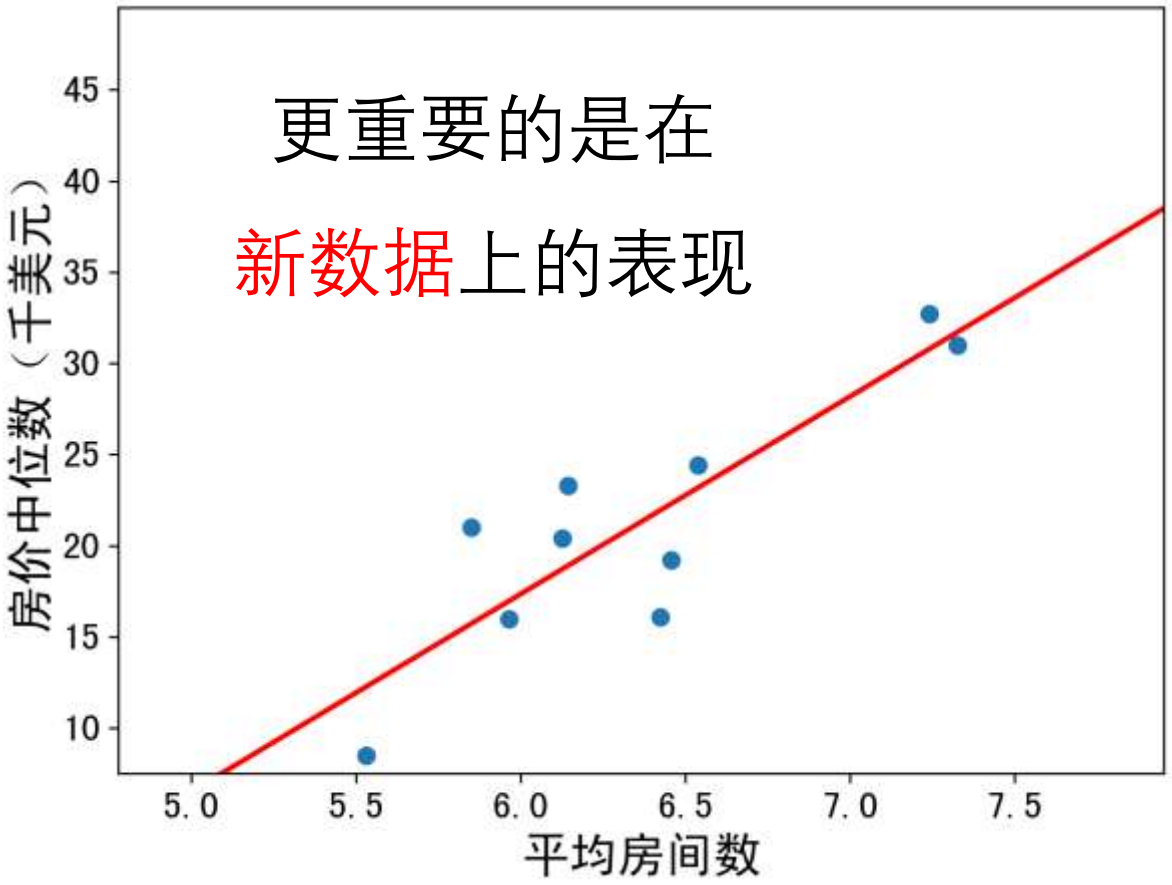
$$y = wx + b$$

- 梯度下降拟合模型

Step 3: 测试训练结果

$b = -0.2283 \quad w = 0.8276$

$error = 2.8898$



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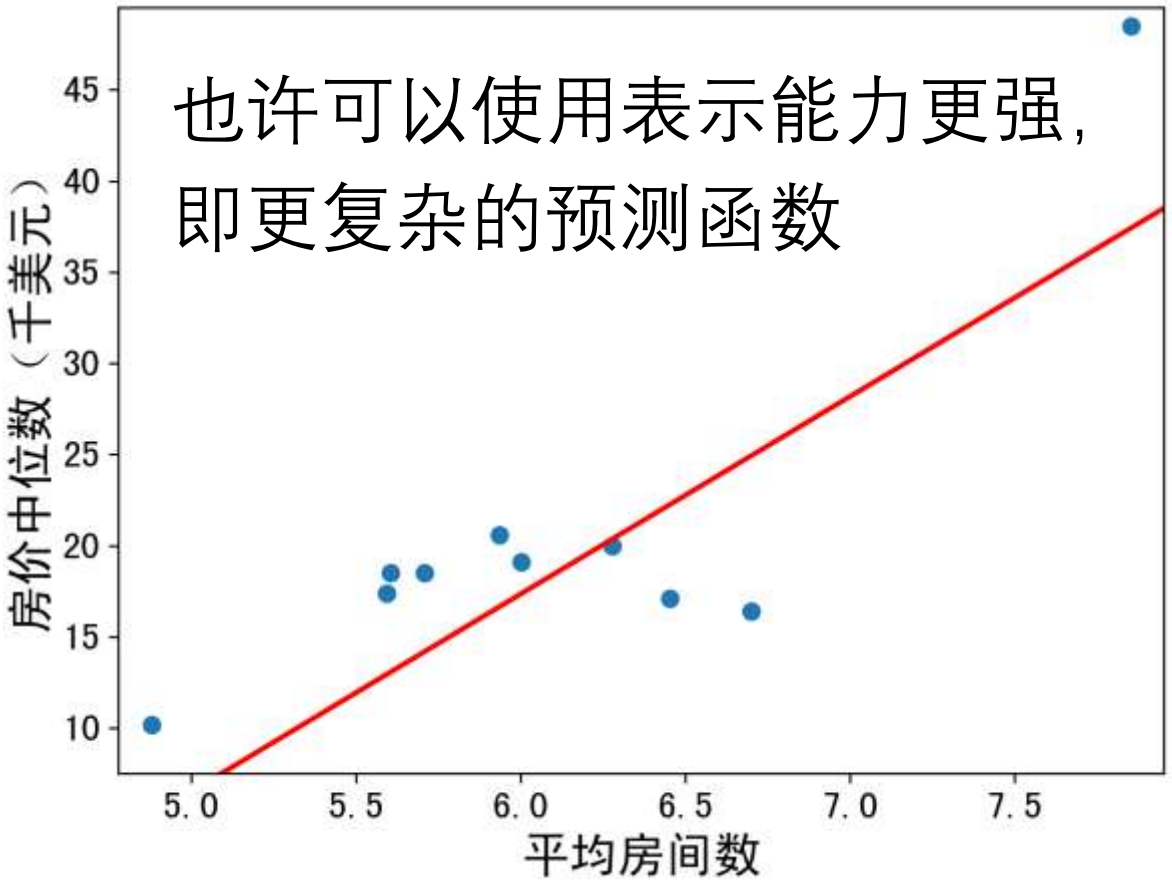


$$test_error = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i| = 4.9989$$

泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1 x + w_2 x^2$$

最优的模型

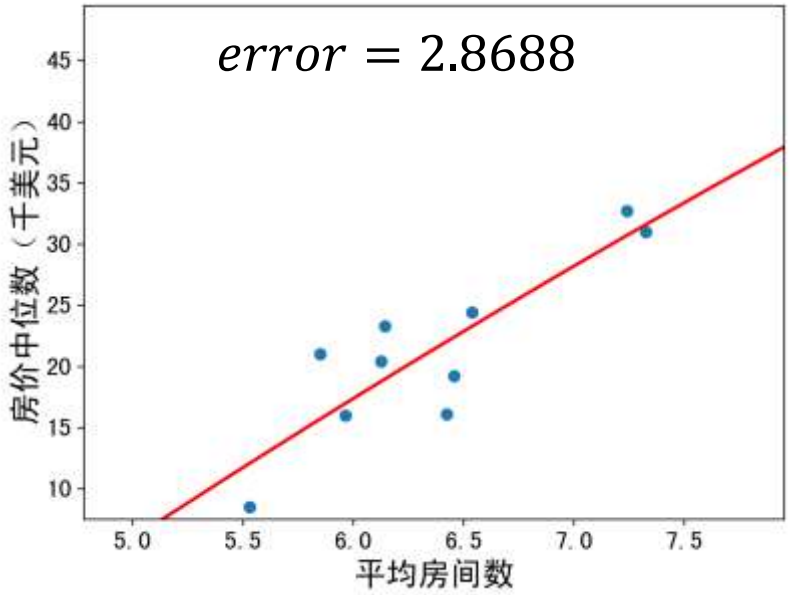
$b = -0.2283$

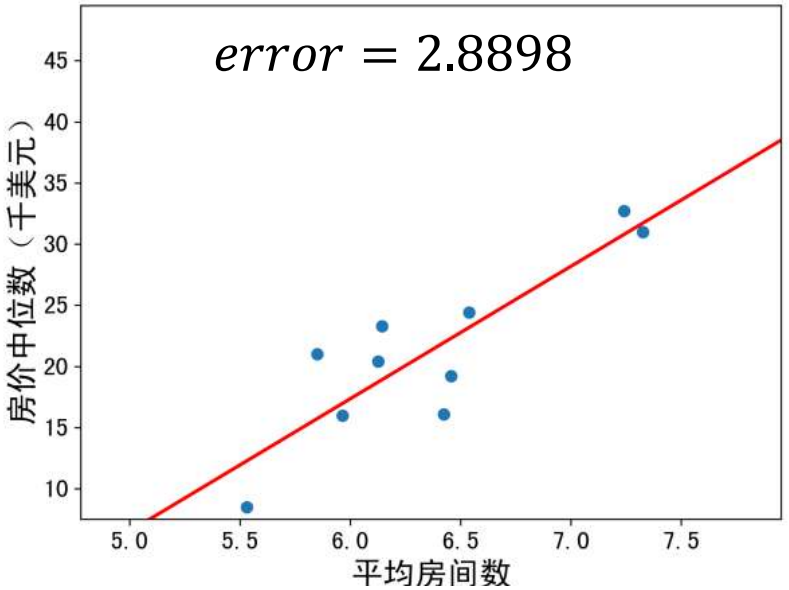
$w_1 = 0.8377, w_2 = -0.0167$

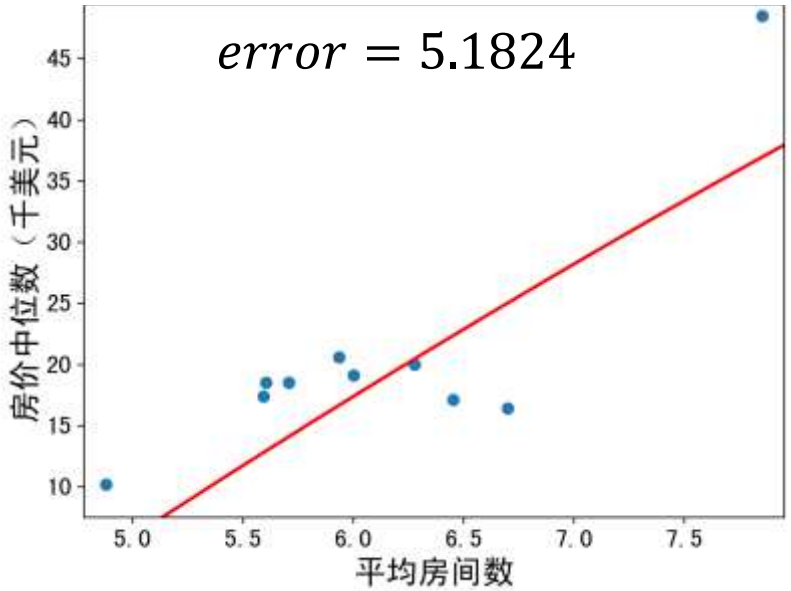
测试结果

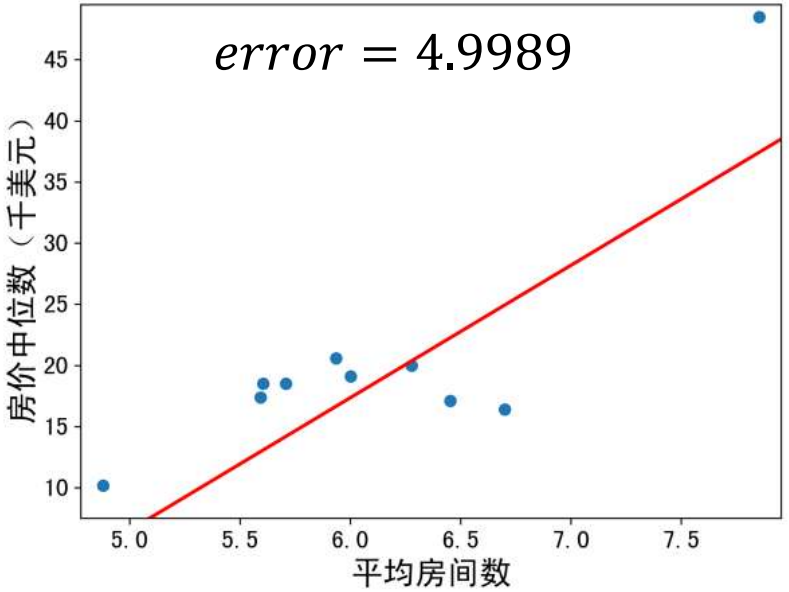
$train_error = 2.8688$

$test_error = 5.1824$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1 x + w_2 x^2 + w_3 x^3$$

最优的模型

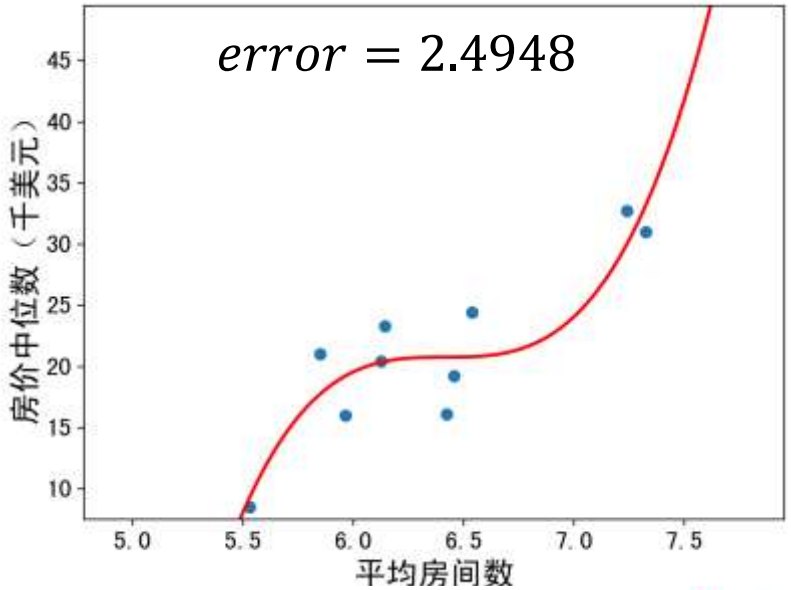
$b = -0.1980$

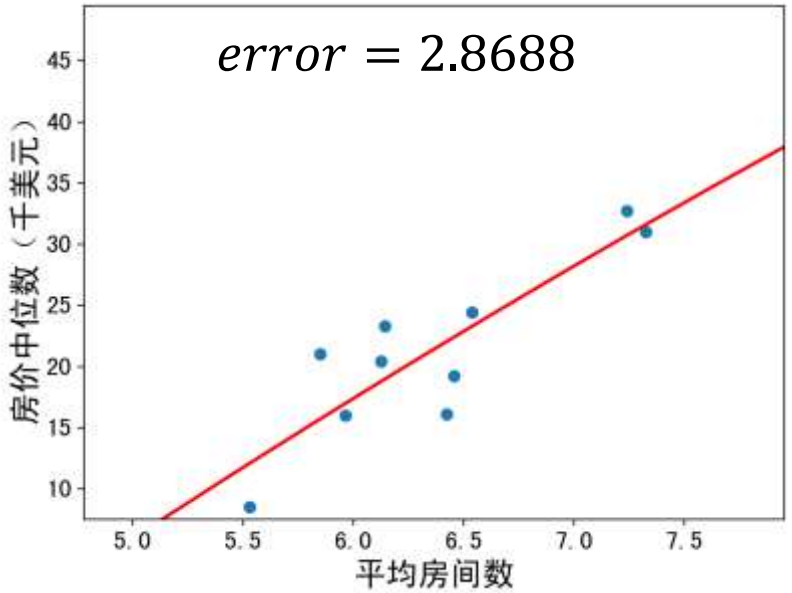
$w_1 = 0.0689, w_2 = -0.3491,$
$w_3 = 0.6201$

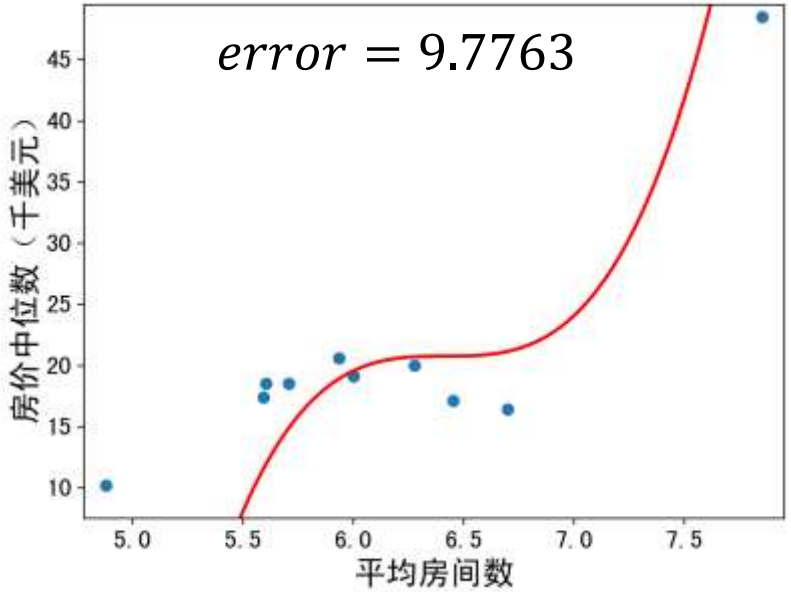
测试结果

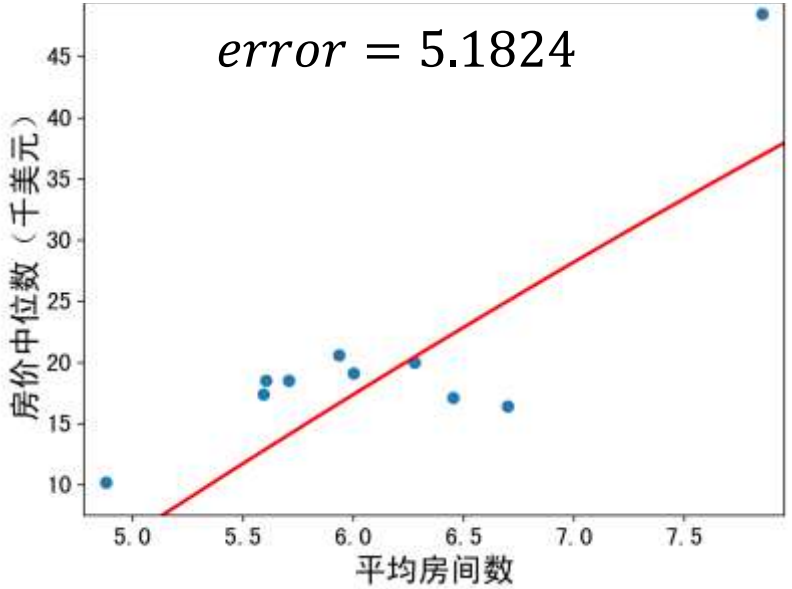
$train_error = 2.4948$

$test_error = 9.7763$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4$$

最优的模型

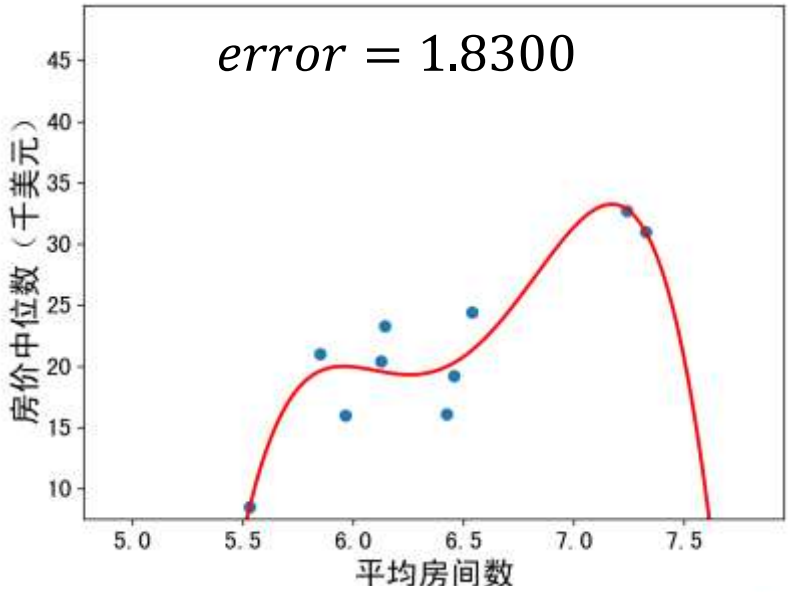
$b = -0.3477$

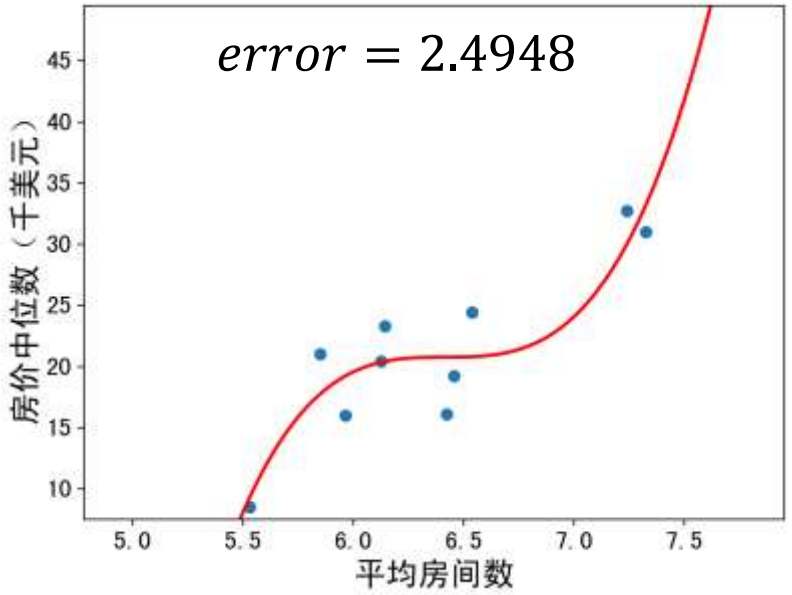
$w_1 = 0.0915, w_2 = 1.1840,$
$w_3 = 0.9572, w_4 = -0.9501$

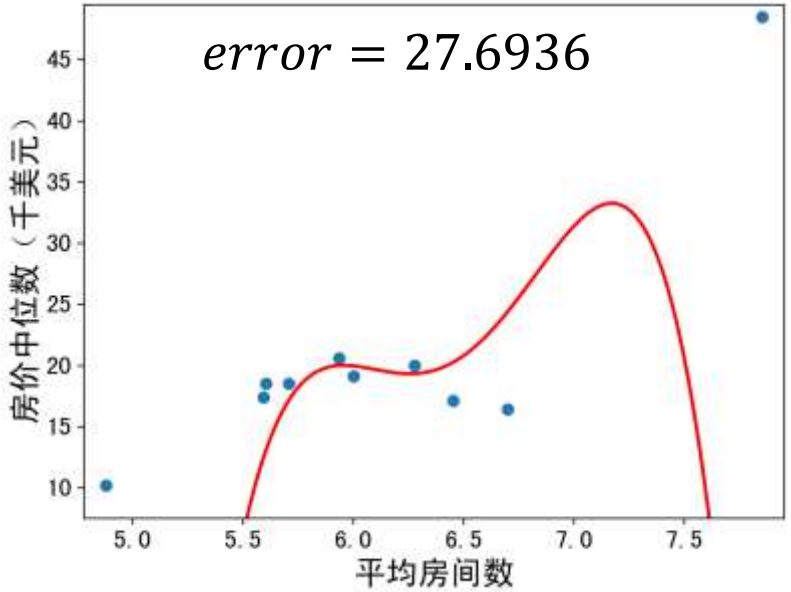
测试结果

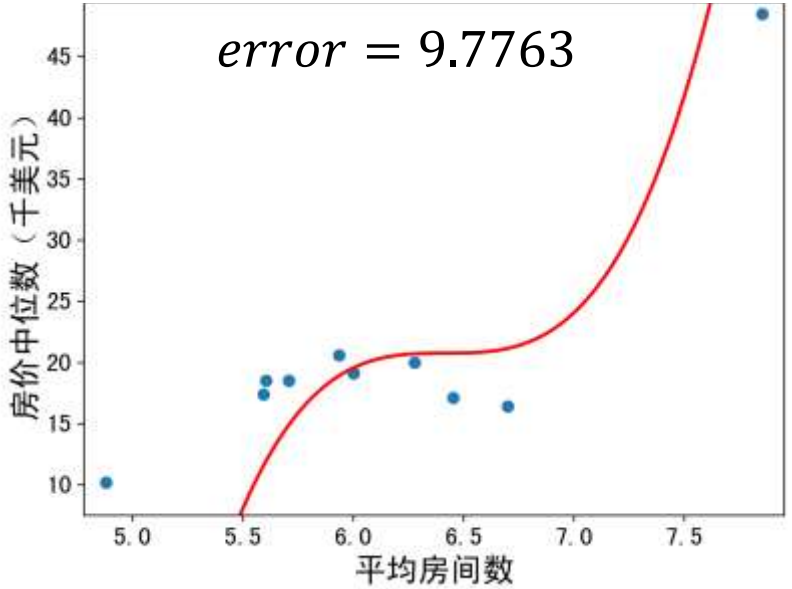
$train_error = 1.830$

$test_error = 27.6936$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5$$

最优的模型

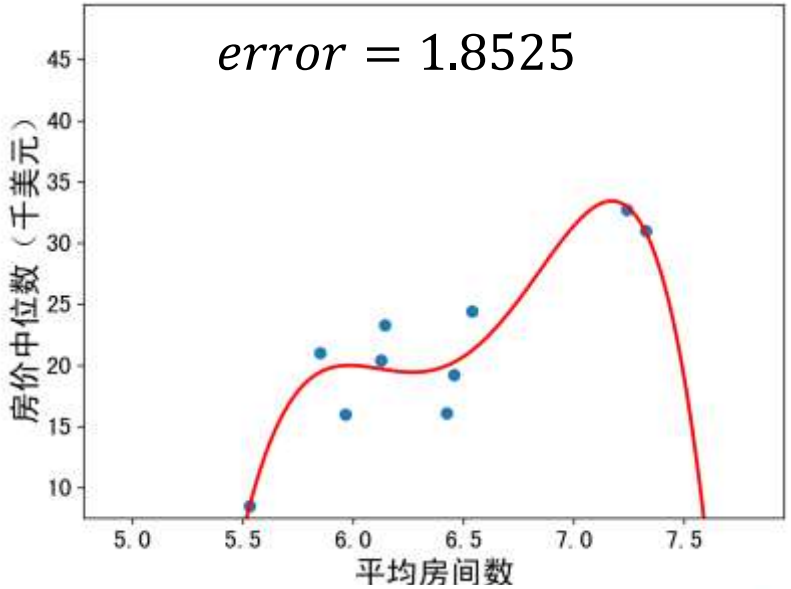
$b = -0.3361$

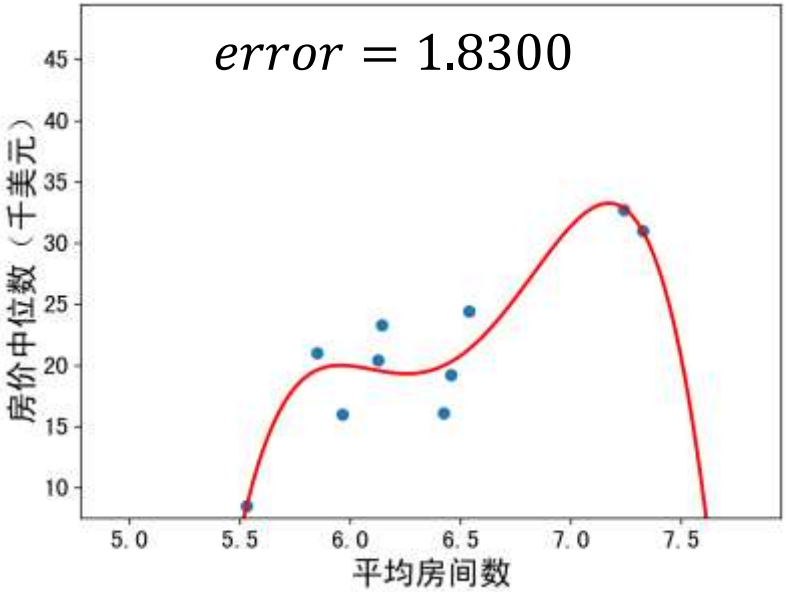
$w_1 = 0.0479, w_2 = 1.0634,$
$w_3 = 1.1011, w_4 = -0.8494,$
$w_5 = -0.0904$

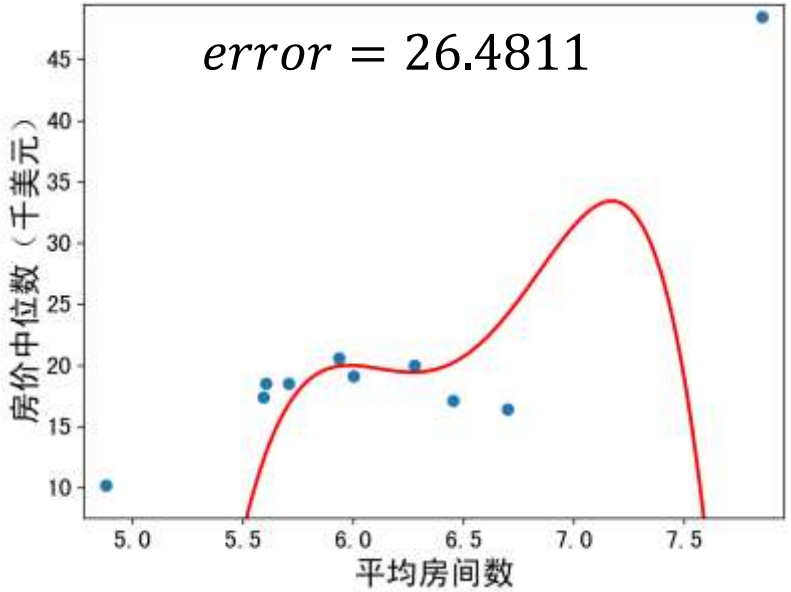
测试结果

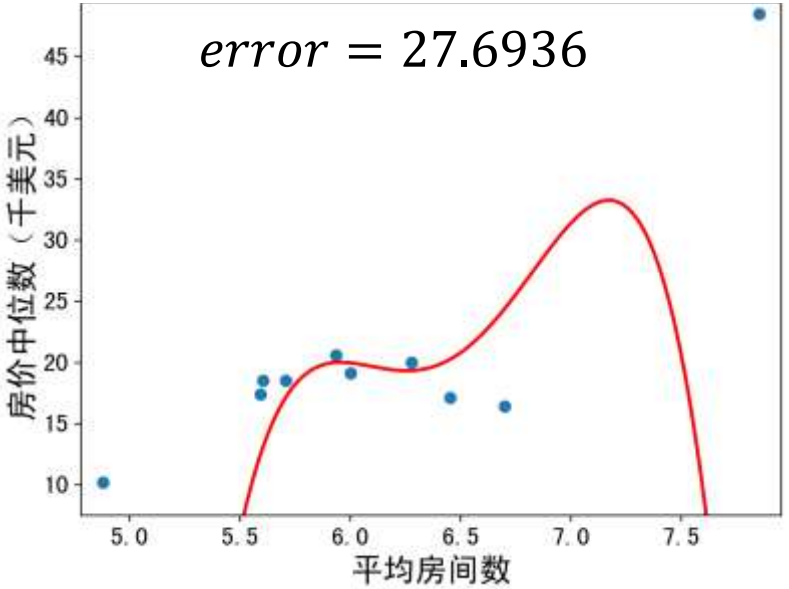
$train_error = 1.8525$

$test_error = 26.4811$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5 + w_6 x^6$$

最优的模型

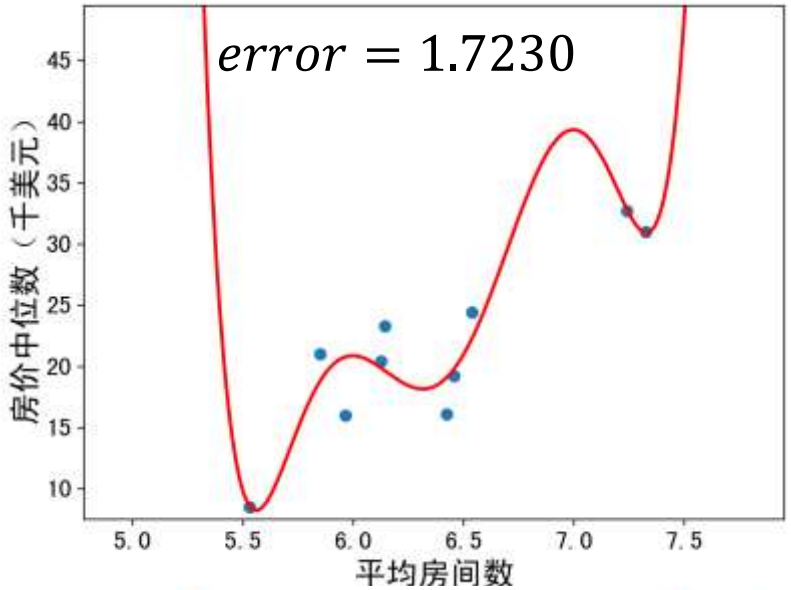
$b = -0.4694$

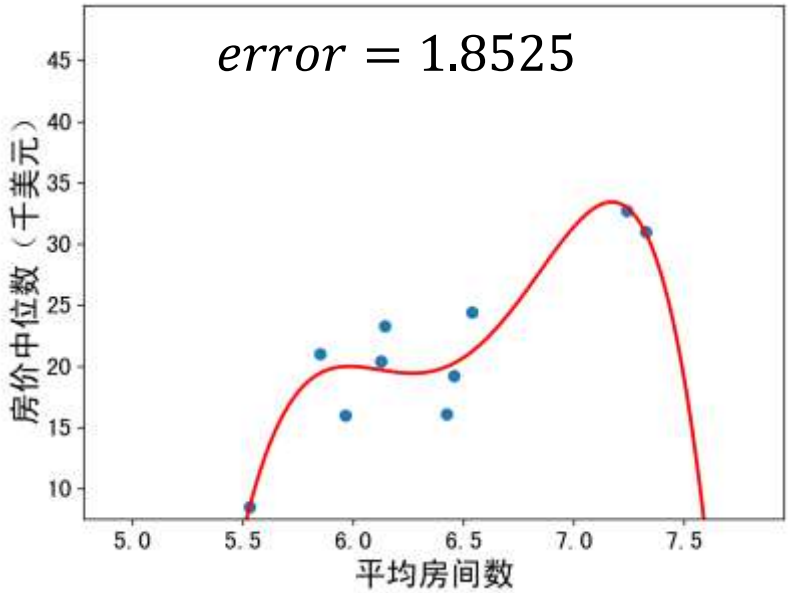
$w_1 = -0.3189, w_2 = 3.3324,$
$w_3 = 4.5095, w_4 = -4.6067,$
$w_5 = -2.5075, w_6 = 1.8883$

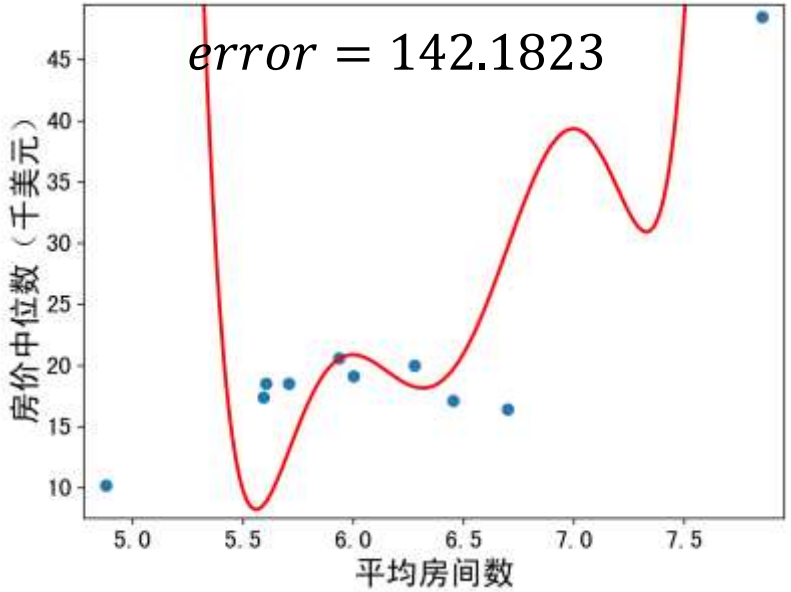
测试结果

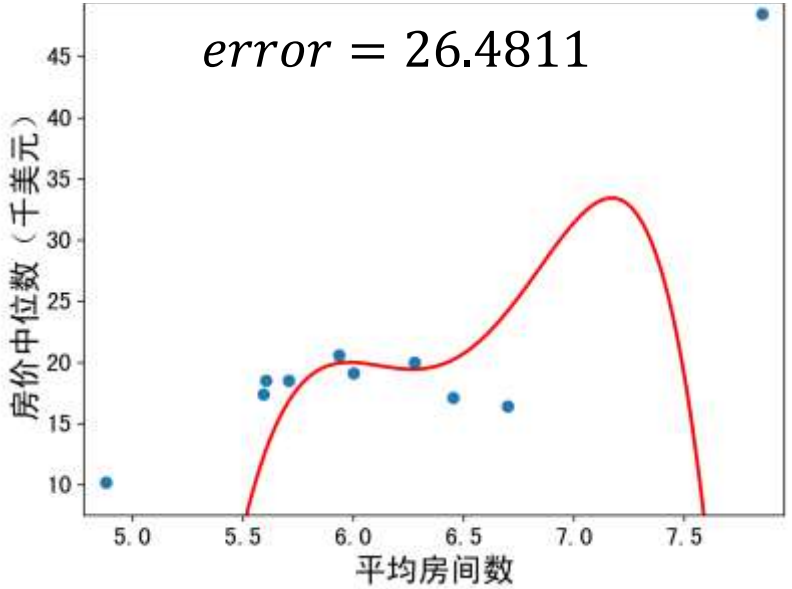
$train_error = 1.7230$

$test_error = 142.1823$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7$$

最优的模型

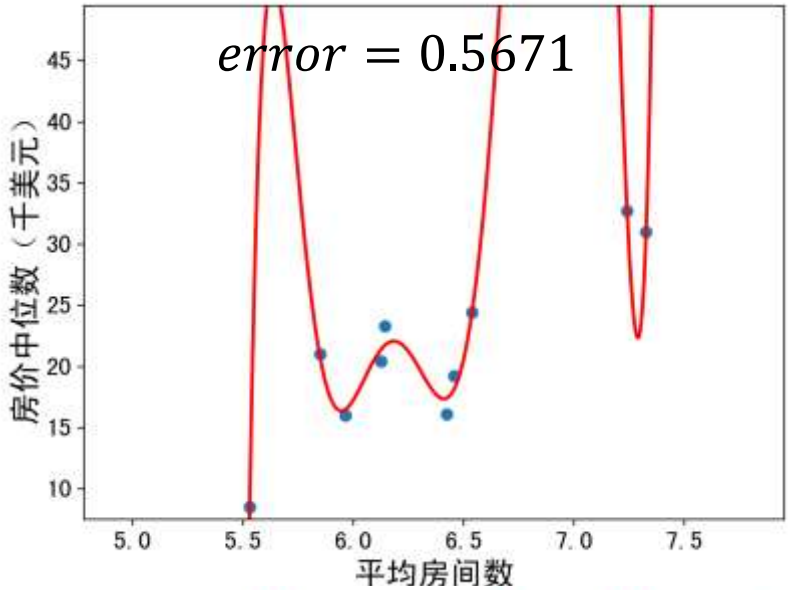
$b = -0.2436$

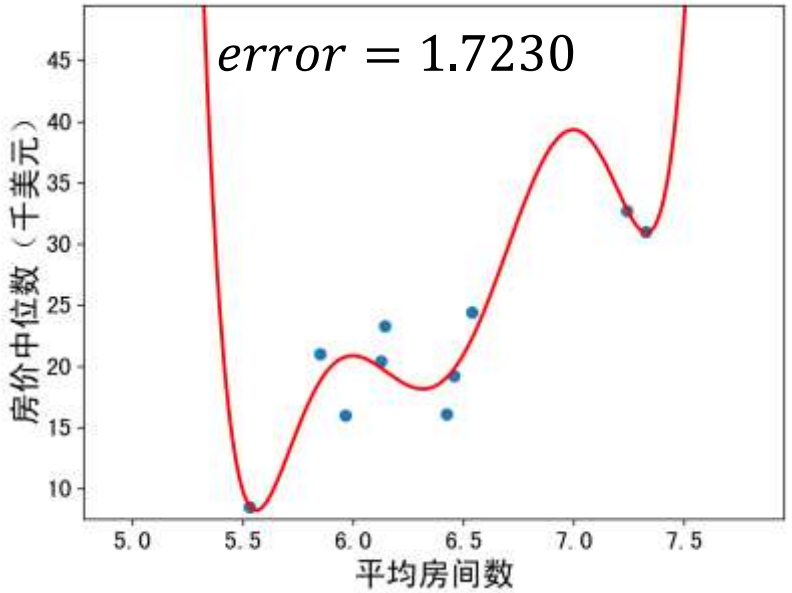
$w_1 = -2.3153, w_2 = -3.3200,$
$w_3 = 30.2344, w_4 = 35.3646,$
$w_5 = -43.9501, w_6 = -25.0806$
$w_7 = 20.5516$

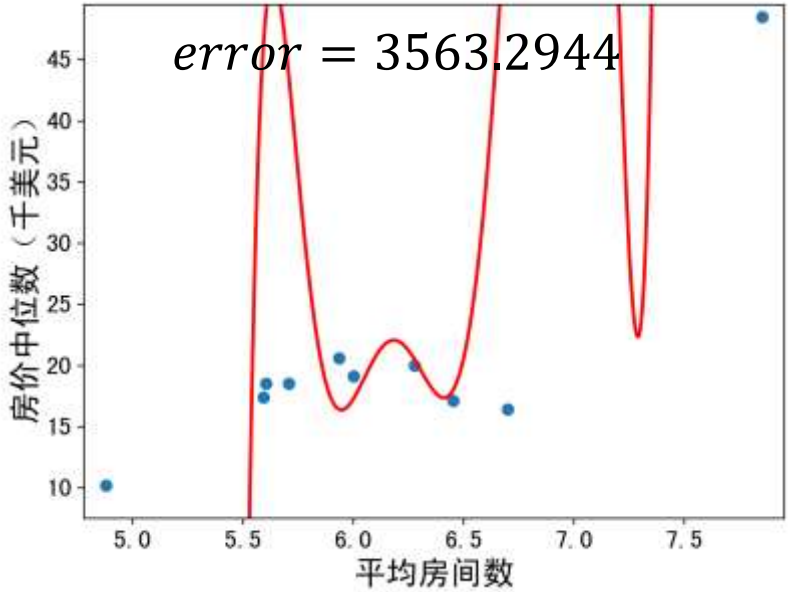
测试结果

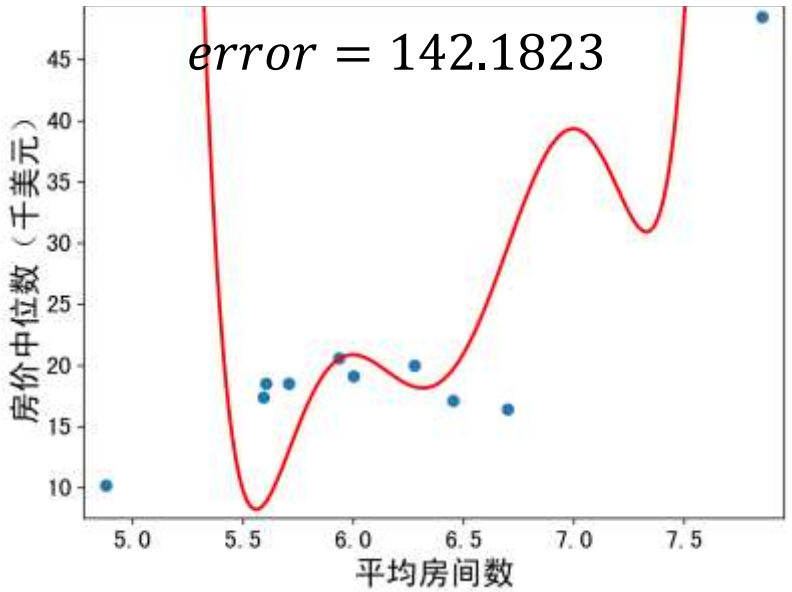
$train_error = 0.5671$

$test_error = 3563.2944$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8$$

最优的模型

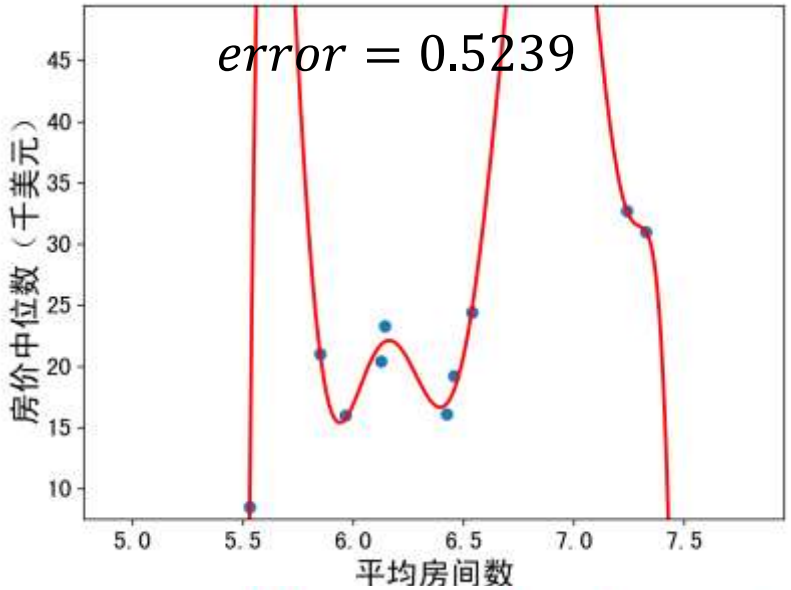
$b = -0.3537$

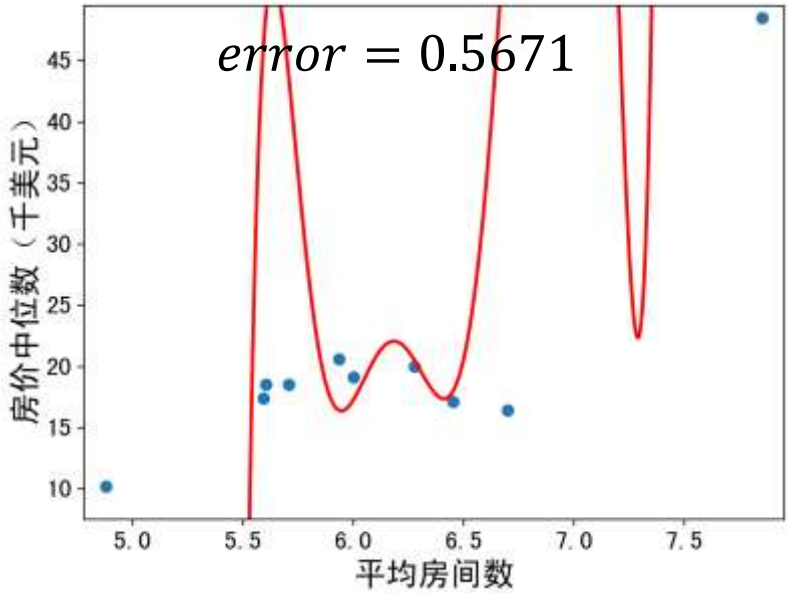
$w_1 = -2.7625, w_2 = 0.0784,$
$w_3 = 37.6561, w_4 = 12.7542,$
$w_5 = -75.0720, w_6 = -7.5115,$
$w_7 = 40.2999, w_8 = -15.2368$

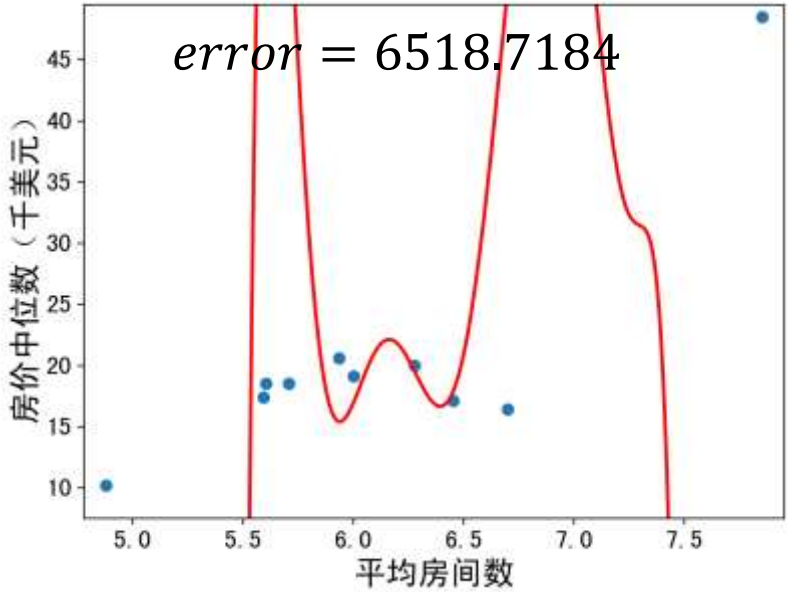
测试结果

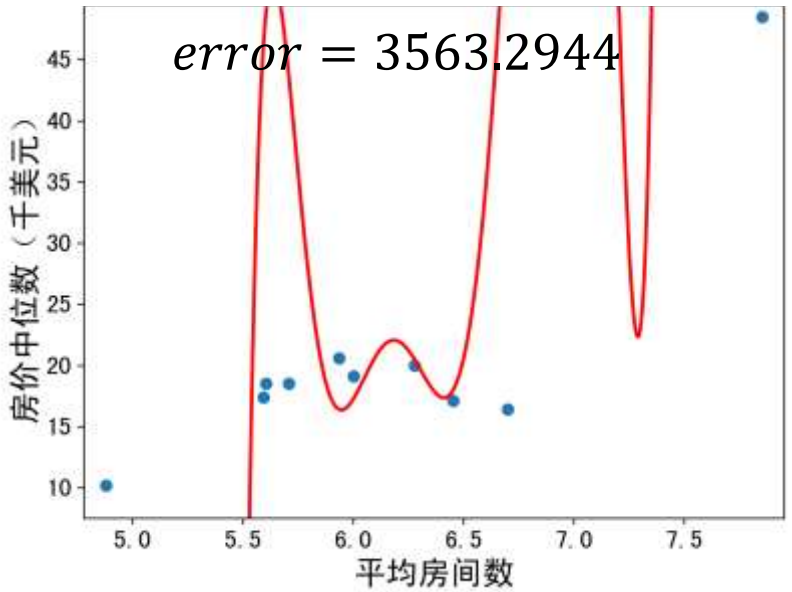
$train_error = 0.5239$

$test_error = 6518.7184$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9$$

最优的模型

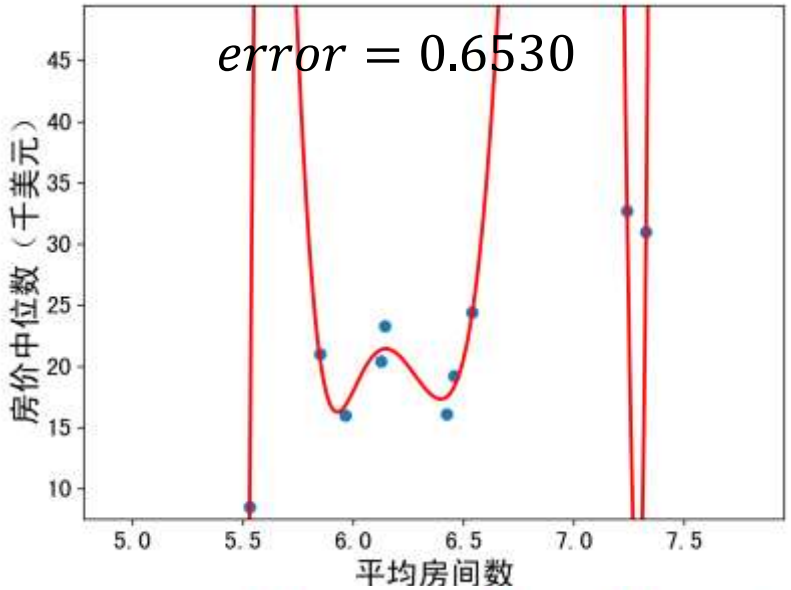
$b = -0.3663$

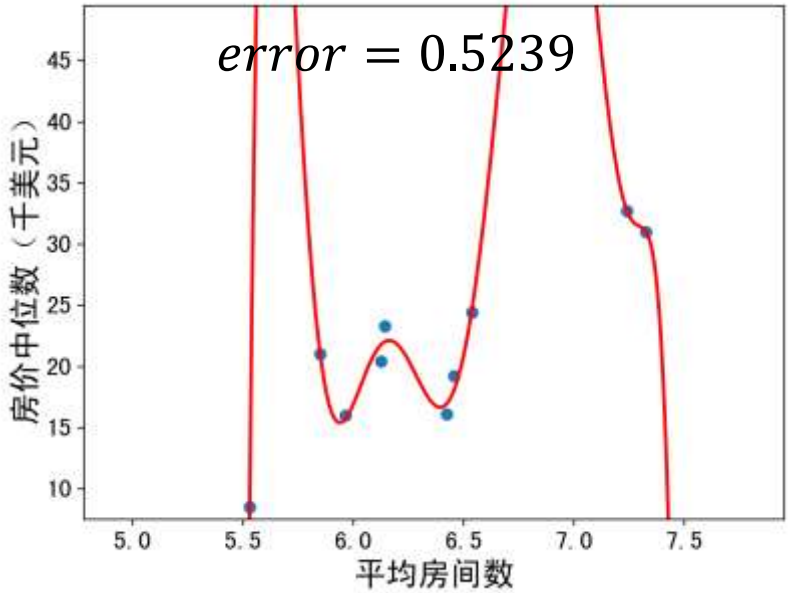
$w_1 = -1.8967, w_2 = 0.5591,$
$w_3 = 20.7009, w_4 = 5.8272,$
$w_5 = 3.5024, w_6 = 51.2704,$
$w_7 = -46.0448, w_8 = -43.2955,$
$w_9 = 31.0031$

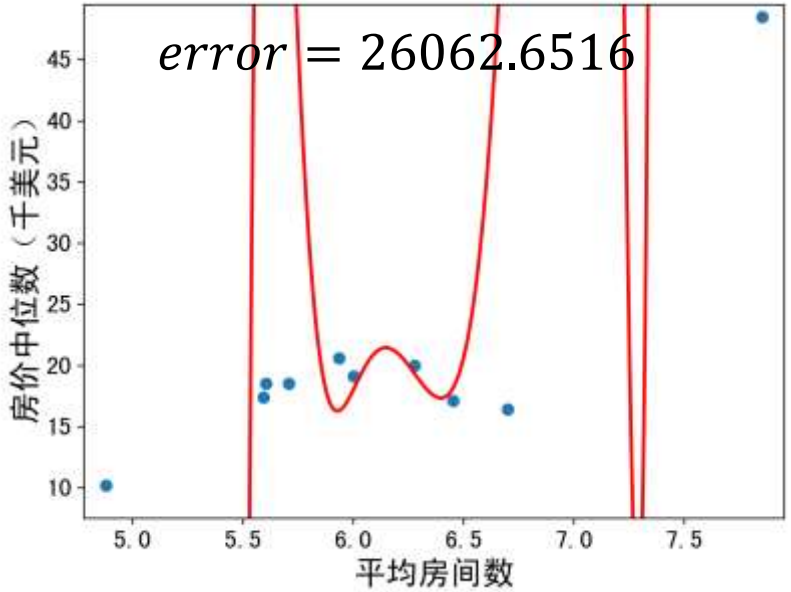
测试结果

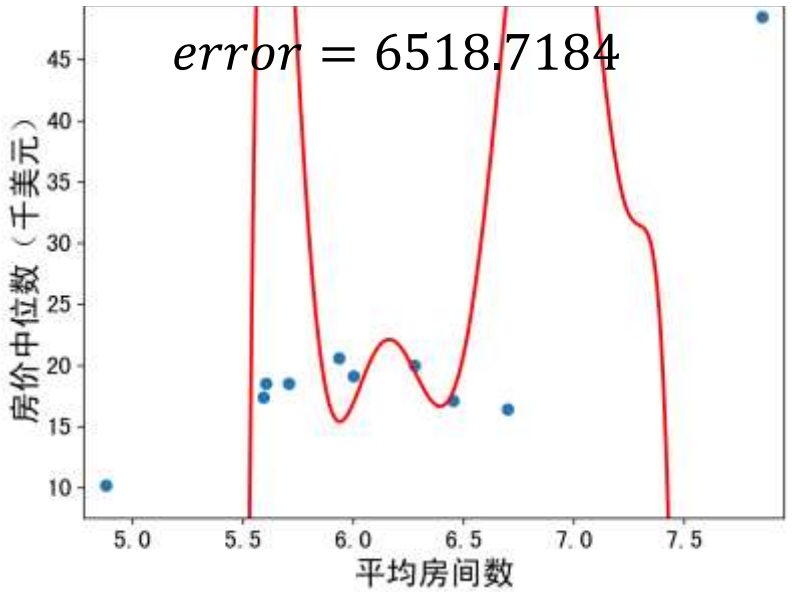
$train_error = 0.6530$

$test_error = 26062.6516$









泛化能力评估——实际案例

更复杂的预测模型

$$y = b + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8 + w_9x^9 + w_{10}x^{10}$$

最优的模型

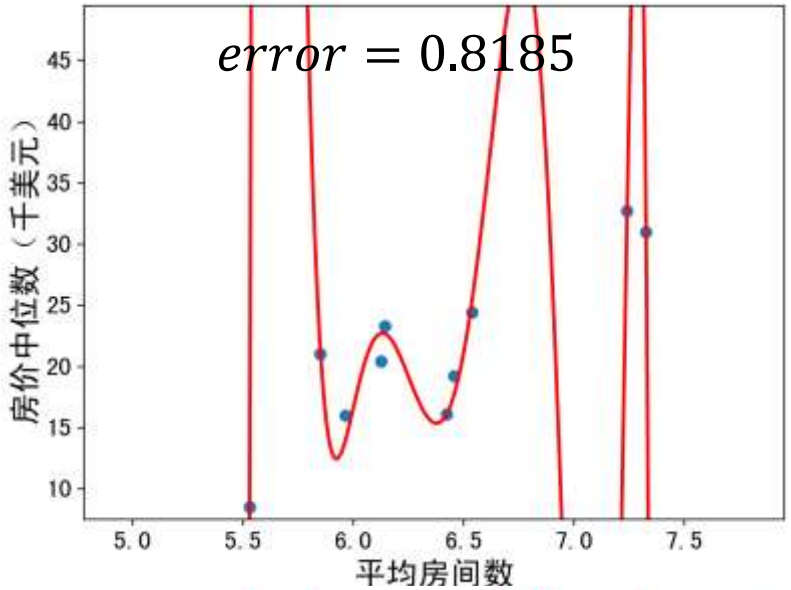
$b = -0.5319$

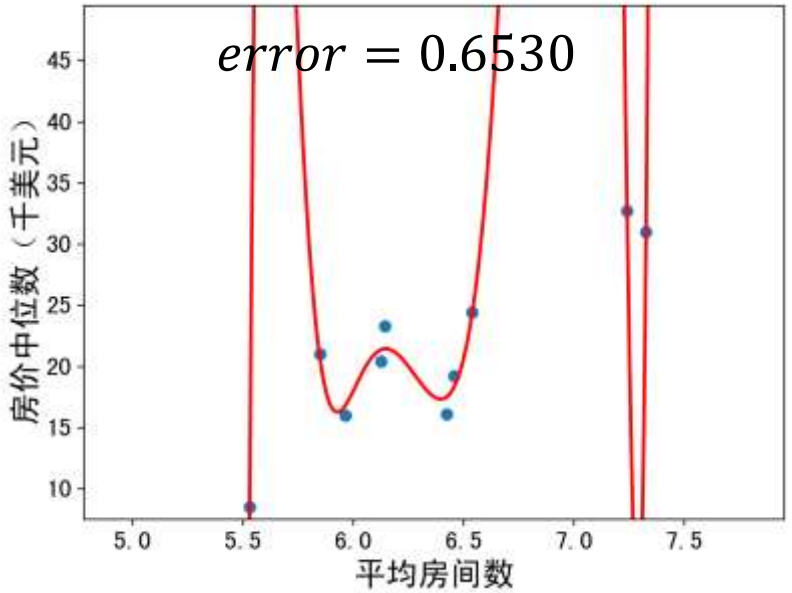
$w_1 = -3.2218, w_2 = 5.5576,$
$w_3 = 37.4427, w_4 = -21.0840,$
$w_5 = -26.8382, w_6 = 53.4348,$
$w_7 = -120.6072, w_8 = 18.0439,$
$w_9 = 100.8242, w_{10} = -47.0201$

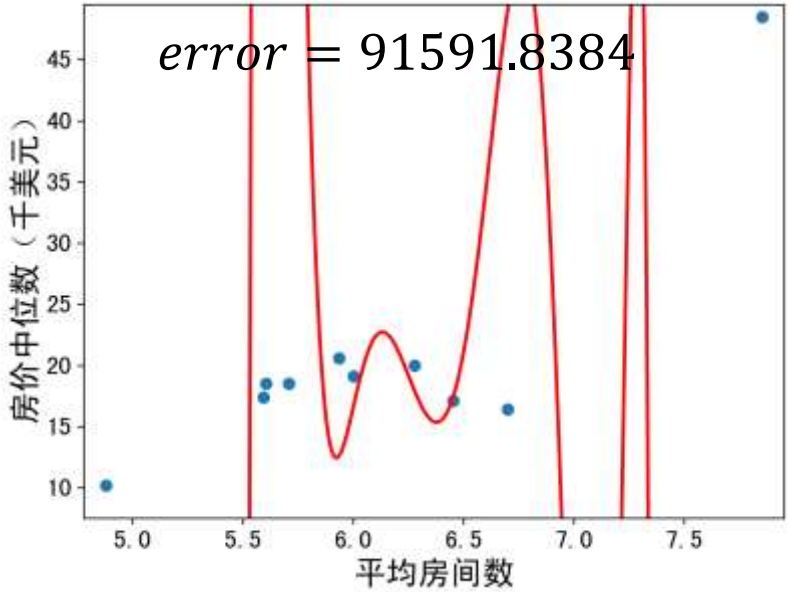
测试结果

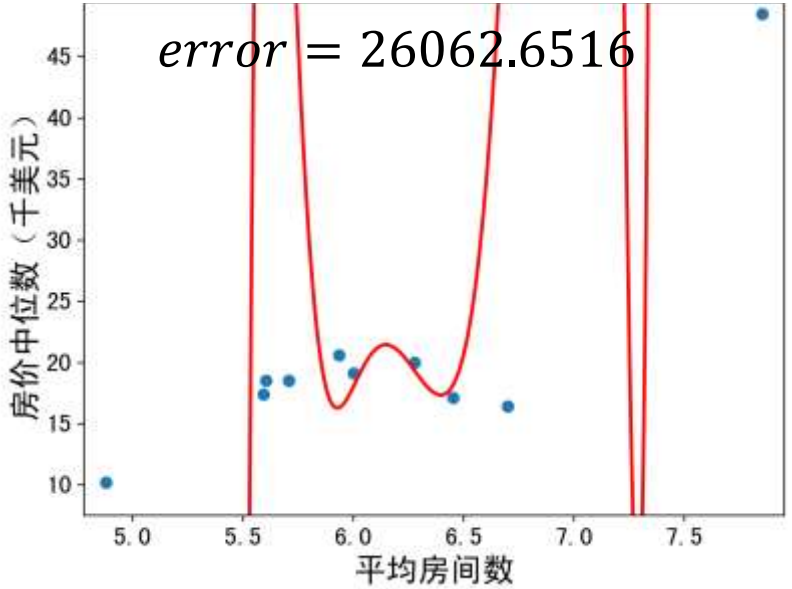
$train_error = 0.8185$

$test_error = 91591.8384$









泛化能力评估——实际案例

预测模型

$y = b + w_1 x$

$y = b + w_1 x + w_2 x^2$

$y = b + w_1 x + w_2 x^2 + w_3 x^3$

$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4$

$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5$

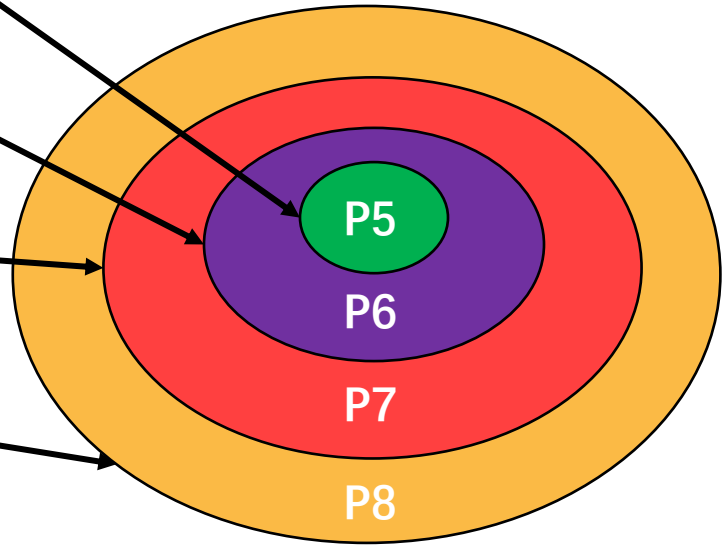
$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5 + w_6 x^6$

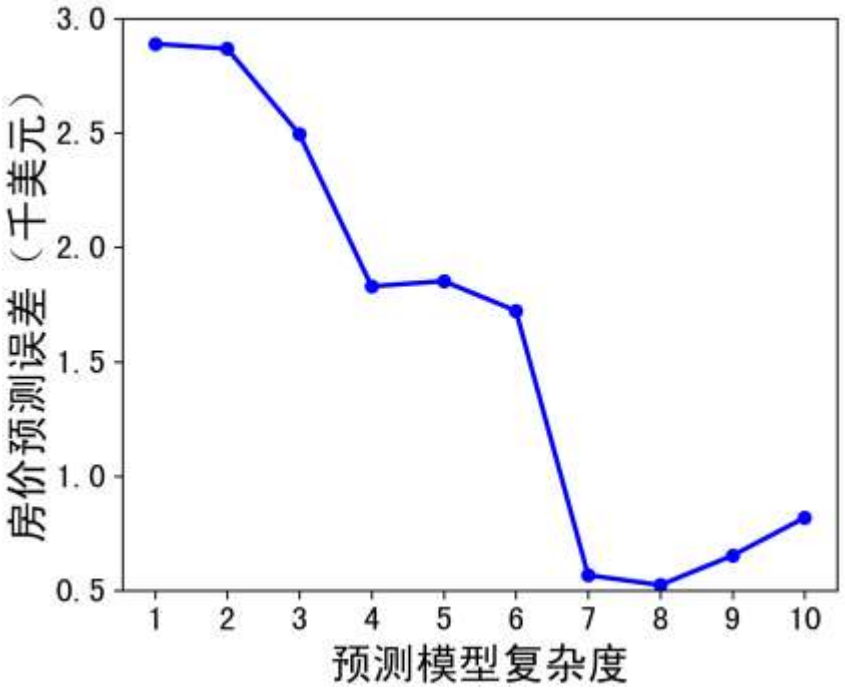
$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5 + w_6 x^6 + w_7 x^7$

$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5 + w_6 x^6 + w_7 x^7 + w_8 x^8$

$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5 + w_6 x^6 + w_7 x^7 + w_8 x^8 + w_9 x^9$

$y = b + w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + w_5 x^5 + w_6 x^6 + w_7 x^7 + w_8 x^8 + w_9 x^9 + w_{10} x^{10}$



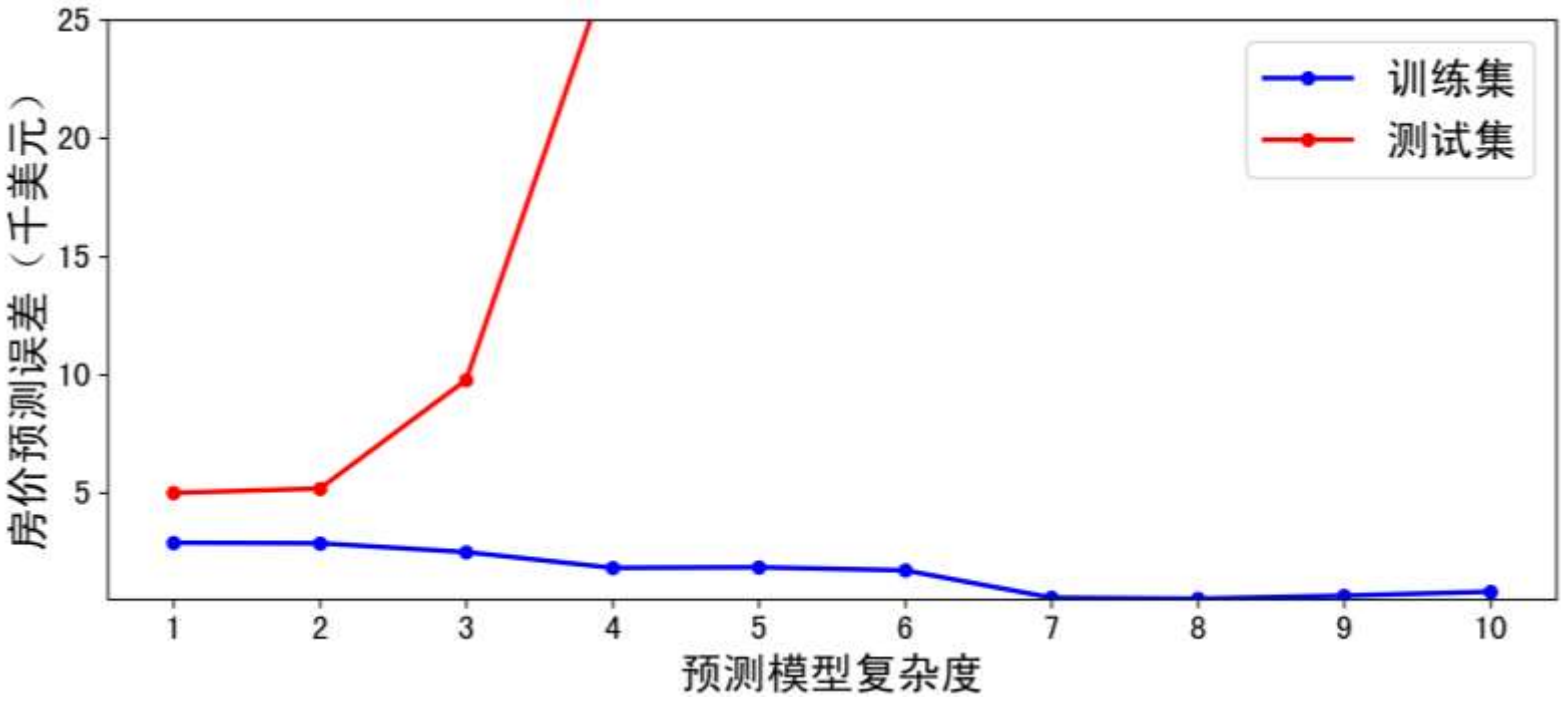


复杂的模型具有较强的表示能力，能够捕捉到更加细节的特征。

泛化能力评估——实际案例

预测模型的泛化结果

复杂度	训练误差	测试误差
1	2.8898	4.9989
2	2.8688	2.1824
3	2.4948	9.7763
4	1.8300	27.6936
5	1.8525	26.4811
6	1.7230	142.1823
7	0.5671	3563.2944
8	0.5238	6518.7184
9	0.6230	26062.6516
10	0.8185	91591.8384



复杂的模型不一定有更好的表现，可能过度关注训练数据的细节，丧失了对一般问题的理解。这就是传说中的**过拟合**问题。

一个发生了过拟合的人

极其熟悉训练环境与操作的细节

无法直接在新环境中顺利完成入库

通过放置水桶还原训练场景中的关键参考物，不失为一种泛化方法。

思考并讨论：有哪些方法可以让此人在训练过程中发生过拟合？